

ASTHA AHIRWAR

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Summary

Mathematics and Computing undergraduate with hands-on experience building machine learning and full-stack applications, including ML-powered dashboards, fraud detection systems, and relational-database-driven web apps. Proficient in Python, C++, and SQL, with a strong foundation in data structures, DBMS, and applied statistics. Seeking opportunities to apply analytical and software development skills to solve real-world problems.

Education

Sardar Vallabhbhai National Institute of Technology, Surat
Bachelor of Technology in Mathematics and Computing

July 2024 – Present
CGPA: 7.73/10

Skill Summary

Languages: C++, Python, SQL (MySQL, Oracle)

Technologies: NumPy, Pandas, Matplotlib, Git, GitHub, VS Code, Streamlit, Flask

Concepts: Data Structures, Object Oriented Programming, Data Base Management System, Linear Algebra, Probability, Discrete Mathematics

Projects

Carbon Emission Prediction and Scenario Simulation for Villages *Python, Streamlit, Scikit-learn, XGBoost, SHAP*

- Developed an ML-powered dashboard to predict household carbon emissions using Linear Regression, Random Forest, and XGBoost models.
- Performed automated EDA, feature engineering, model evaluation, and SHAP-based explainability for model interpretation.
- Built an interactive Streamlit application for real-time sustainability scenario analysis and emission forecasting.

Credit Card Fraud Detection System

Python, Scikit-learn, Random Forest, SMOTE, Pandas

- Built a fraud detection model for highly imbalanced financial transaction data using Random Forest and SMOTE.
- Achieved **99.94% accuracy**, **89% precision**, and **78% recall** through data preprocessing and model optimization.
- Implemented a real-time transaction classification pipeline and evaluated performance using Precision, Recall, and F1-Score metrics.

Smart Parking Management System

Flask, MySQL, HTML, CSS, Bootstrap

- Developed a full-stack parking management application with user authentication, slot booking, booking history, and payment tracking.
- Designed a relational MySQL database and implemented MVC-based OOP architecture for efficient data management.
- Built dynamic dashboards with real-time slot availability updates and automated parking charge calculation.

Relevant Coursework

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|-------------------------------|-------------------------------|
| • Data Structures | • Object-Oriented Programming |
| • Database Management Systems | • Probability and Statistics |
| • Algorithm Analysis | • Linear Algebra |
| • Computer Networks | • Discrete Mathematics |